# Maksymilian Siekliński

+44 0770-763-1688 | +48 570-383-665 | maksymilian.sieklinski.2@gmail.com

## Education

**University College London** Sept. 2024 – June 2027
*Bachelor of Science, Computer Science (86.3% total avg)* *Great Britain, London*

**V High School in Cracow** Sept. 2020 – Apr. 2024
*GPA 5.55, Top 1% in Math/Physics/CS ($A^*A^*A^*$ equiv.)* *Poland, Cracow*

## Projects

**Carbon optimized AI workload scheduling *(C++, Drogon)*** Sept. 2025 - Mar. 2026

- Developed a C++23 web app to minimize AI carbon emissions via global datacenter scheduling.
- Optimized workload execution using dynamic programming to select the greenest time and location.
- Implemented a lock-free architecture to manage shared state across high-concurrency threads.
- Vectorized performance-critical code using AVX-512 for SIMD parallelization.
- Leveraged C++23 coroutines and engineered a custom *when_all* asynchronous primitive.

**Verified Market Simulator Research *(Python, OCaml)*** Jan. 2026 - Present
*Ongoing*

- Architecting a purely functional LOB simulator in OCaml 5.0 using immutable structures to eliminate state drift.
- Implementing a low-latency streamer using TCP sockets and zero-copy parsing.

## Experience

**IMC Trading — Software Engineering Internship** Jun. 2026 - Sept. 2026
*Execution Team*

- Architected a C++ load-testing framework capable of generating 13M+ updates/sec across burst traffic profiles.
- Benchmarked FPGA-accelerated execution gateway to establish 6M updates/sec as the maximum throughput.
- Uncovered gateway crash vectors in session handling, client disconnects, and ID counter overflows.
- Eliminated firm-wide tail-latency spikes by resolving unbounded hash map growth to prevent runtime rehashing.
- Automated CI/CD performance regression suites to verify throughput and stability on code changes.

**Huawei — Programming Language Research Intern *(Cangjie)*** Jun. 2025 - Sept. 2025

- Built an AI-driven, cross-platform browser automation tool utilizing computer vision for visual classification.
- Packaged in Docker for OS- and CPU-agnostic deployment, simplifying environment setup.

**Insight Programs — IMC Trading, Citadel & Susquehanna** 2025 & 2026

- 1st place in Citadel's quantitative research challenge, a team-based probability competition.
- 1st place in IMC internship automated trading hackathon.
- 3rd place in IMC algorithmic optimization challenge.
- 3rd place in SIG trading challenge.

## Competitions & Awards

- 5th in Poland in group competitive programming contest. 2023
- 8th in Poland in solo competitive programming contest, Diamentowy Index AGH. 2024
- 11th in UK & Ireland Programming Contest for University students (UKIEPC). 2025
- 27th in UK based on Codeforces ranking. Present
- Top 50 in Northwestern Europe Regional Contest (NWERC). 2024, 2025
- Distinction in National Polish Computer Science Olympiad. 2023, 2024

## Skills & Leadership

Founder & President of the UCL Competitive Programming Society, and 3-year Team Captain.

Programming Languages: Advanced C++ and Java, C, Haskell, Ocaml, Python, SQL and Bash.

Algorithmic knowledge: Centroid Decomposition, Heavy-Light Decomposition, FFT & Aho-Corasick.